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Extruded Polystyrene Processing Device

Abstract

An extruded polystyrene processing device for processing extruded polystyrene includes a processing plant which includes a chopping unit for chopping extruded polystyrene into chopped pieces. The chopping unit is connected to a solvent bath unit for submerging the chopped pieces in a solvent for eliminating air bubbles in the chopped pieces. The solvent bath unit is connected to a conveyor unit for rinsing the chopped pieces and the conveyor unit is connected to a freezing unit to subsequently freeze the chopped pieces. The freezing unit is connected to a grinding unit for grinding the frozen chopped pieces into a granular material for subsequent collection and transportation to a waste management facility. In this way the processing plant reduces the total volume of the extruded polystyrene prior to being handled by the waste management facility.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION

(1) Field of the Invention

[0006] The disclosure relates to polystyrene processing devices and more particularly pertains to a new polystyrene processing device for reducing the total volume of extruded polystyrene prior to being handled by a waste management facility. The device includes a processing plant which includes a chopping unit for chopping items of extruded polystyrene into chopped pieces and a solvent bath unit for reducing the chopped pieces and a conveyor unit for rinsing the chopped pieces and a freezing unit for freezing the chopped pieces and a grinding unit for grinding the chopped pieces into a granular material.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

[0007] The prior art relates to polystyrene processing devices including a variety of polystyrene processing devices that mechanically process polystyrene into a crushed product and a polystyrene processing device that employs a solvent to reduce polystyrene into a sludge and a variety of polystyrene processing devices that each at least employs a solvent for reducing the volume of polystyrene. In no instance does the prior art disclose a polystyrene processing device that includes a chopper for chopping items of extruded polystyrene and a solvent bath for reducing the chopped extruded polystyrene and a grinding unit for grinding the reduced polystyrene into a granular material to reduce the overall volume of the polystyrene.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above by generally comprising a processing plant which includes a chopping unit for chopping extruded polystyrene into chopped pieces. The chopping unit is connected to a solvent bath unit for submerging the chopped pieces in a solvent for eliminating air bubbles in the chopped pieces. The solvent bath unit is connected to a conveyor unit for rinsing the chopped pieces and the conveyor unit is connected to a freezing unit to subsequently freeze the chopped pieces. The freezing unit is connected to a grinding unit for grinding the frozen chopped pieces into a granular material for subsequent collection and transportation to a waste management facility. In this way the processing plant reduces the total volume of the extruded polystyrene prior to being handled by the waste management facility.

[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0011] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. **1** is a top perspective view of an extruded polystyrene processing device according to an embodiment of the disclosure.

[0013] FIG. **2** is a left side phantom view of an embodiment of the disclosure.

[0014] FIG. **3** is a magnified detail view taken from circle **3** of FIG. **2** of an embodiment of the disclosure showing a water lift plate in a second position.

[0015] FIG. **4** is a magnified phantom view taken from circle **4** of FIG. **2** of an embodiment of the disclosure showing chopped pieces being deposited into a solvent bath unit.

[0016] FIG. **5** is a magnified phantom view taken from circle **5** of FIG. **2** of an embodiment of the disclosure showing a solvent lift panel being urged into a vertical position.

[0017] FIG. **6** is a top view of a freezer unit of an embodiment of the disclosure.

[0018] FIG. **7** is a top perspective view of a tray of an embodiment of the disclosure.

[0019] FIG. **8** is a logic tree view of an embodiment of the disclosure.

[0020] FIG. **9** is a schematic view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0021] With reference now to the drawings, and in particular to FIGS. **1** through **9** thereof, a new polystyrene processing device embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral **10** will be described.

[0022] As best illustrated in FIGS. **1** through **9**, the extruded polystyrene processing device **10** generally comprises a processing plant **12** which includes a chopping unit **14** for chopping items of extruded polystyrene **16** into chopped pieces **18**. The chopping unit **14** is connected to a solvent bath unit **20** for submerging the chopped pieces **18** in a solvent **22** for eliminating air bubbles in the chopped pieces **18**. The solvent bath unit **20** is connected to a conveyor unit **24** which receives the chopped pieces **18** for rinsing the solvent **22** from the chopped pieces **18**. Additionally, the conveyor unit **24** is connected to a freezing unit **26** which receives the chopped pieces **18** to subsequently freeze the chopped pieces **18**. The freezing unit **26** is connected to a grinding unit **28** which receives the heated and dried chopped pieces **18** for grinding the heated and dried chopped pieces **18** into a granular material **30** for subsequent collection and transportation to a waste management facility. In this way the processing plant **12** reduces the total volume of the items of extruded polystyrene **16** prior to being handled by the waste management facility thereby reducing the environmental impact of the extruded polystyrene **16**. Furthermore, the waste management facility may be a landfill, a waste incineration facility or any other kind of waste management facility that commonly deals with both residential and commercial waste.

[0023] The processing plant **12** includes a processing circuit **32** and the processing circuit **32** receives a chop input, a door open input, a door close input, a compress input, a solvent lift input, a conveyor input, a freezing actuate input, a freezing de-actuate input and a grinding input. Additionally, the processing circuit **32** is electrically coupled to a power source **34** which may comprise an electrical system of the facility in which the processing plant **12** is stationed. The chopping unit **14** comprises a hopper **36** which has a first end **38**, a second end **40** and an outer wall **42** extending between the first end **38** and the second end **40**. The hopper **36** is substantially hollow and each of the first end **38** and the second end **40** is open. The outer wall **42** undulates between the first end **38** and the second end **40** such that the hopper **36** defines an S-shape having the first end **38** being spaced upwardly from the second end **40**. In this way the first end **38** can receive items of extruded polystyrene **16** such that the undulation of the outer wall **42** facilitates the items of extruded polystyrene **16** to be gravity fed toward the second end **40**.

[0024] The chopping unit **14** includes a chopper **44** that is movably integrated within the hopper **36** and the chopper **44** is electrically coupled to the processing circuit **32**. The chopper **44** is turned on when the processing circuit **32** receives the chop input and the chopper **44** is positioned closer to the first end **38** than the second end **40**. The chopper **44** includes a series of blades **46** which are rotated when the chopper **44** is turned on thereby facilitating the chopper **44** to chop the items of extruded polystyrene **16** into the chopped pieces **18**. The chopper **44** may comprise an electric motor or the like and the series of blades **46** may be cylindrical blades that are arranged in a row and which are each driven by the electric motor. Additionally, the chopper **44** may be engineered to produce chopped pieces which have a maximum length and width ranging between approximately 5.0 cm and 8.0 cm.

[0025] The chopping unit **14** includes a sensor **48** which is positioned within the hopper **36** and the sensor **48** is positioned closer to the second end **40** of the hopper **36** than the chopper **44**. The sensor **48** is electrically coupled to the processing circuit **32** and the sensor **48** senses the volume of chopped pieces **18** in the hopper **36**. Furthermore, the processing circuit **32** receives the door open input when the sensor **48** determines that the volume of the chopped pieces **18** in the hopper **36** exceeds a pre-determined trigger volume. The pre-determined trigger volume may be a volume of approximately 80.0 percent of the internal area in the hopper **36** which is located between the chopper **44** and the second end **40** of the hopper **36**. Furthermore, the processing circuit **32** receives the door close input when the sensor determines that the volume of the chopped pieces **18** in the hopper **36** does not exceed the pre-determined trigger volume.

[0026] The chopping unit **14** includes a door **50** which is slidably integrated into the second end **40** of the hopper **36** and the door **50** is oriented to lie on a plane that is oriented parallel to the second end **40** of the hopper **36**. The door **50** is positionable in a closed position having the door **50** lying against the second end **40** for closing the second end **40** thereby inhibiting the chopped pieces **18** from passing through the second end **40** and exiting the hopper **36**. The door **50** is positionable in an open position having the door **50** being lifted upwardly to expose the second end **40** thereby facilitating the chopped pieces **18** to pass through the second end **40** to exit the hopper **36**.

[0027] The chopping unit **14** includes a door actuator **52** which is integrated into the hopper **36**. The door actuator **52** is electrically coupled to the processing circuit **32** and the door actuator **52** is in mechanical communication with the door **50**. The door actuator **52** is actuated into a closing condition when the processing circuit **32** receives the door close input thereby urging the door **50** into the closed position. Conversely, the door actuator **52** is actuated into an open condition when the processing circuit **32** receives the door open input thereby urging the door **50** into the open position. The door actuator **52** may be an electric motor or other type of electromechanical actuator that can generate rotational torque in two directions.

[0028] The solvent bath unit **20** comprises a solvent housing **54** that has a front wall **56**, a back wall **58**, a top wall **60** and a bottom wall **62**. The solvent housing **54** has an entrance **64** extending through the back wall **58** and the solvent housing **54** has an exit **66** extending through the front wall **56**; each of the exit **66** and the entrance **64** is spaced upwardly from the bottom wall **62**. The second end **40** of the hopper **36** is attached to the back wall **58** of the solvent housing **54** having the second end **40** being aligned with the entrance **64**. In this way the chopped pieces **18** can pass through the entrance **64** when the door **50** is urged into the open position.

[0029] The solvent housing **54** has a drain **68** which is integrated into the bottom wall **62** thereby facilitating the drain **68** to drain the solvent **22** which is contained in the solvent housing **54**. The solvent **22** is filled to a level that is below each of the entrance **64** and the exit **66** in the solvent housing **54** thereby facilitating the solvent housing **54** to contain the solvent **22**. Additionally, the solvent housing **54** has a spray nozzle **70** extending through the top wall **60** of the solvent housing **54**. The spray nozzle **70** is in fluid communication with the drain **68** thereby facilitating the spray nozzle **70** to spray the solvent **22** into the solvent housing **54** for recirculating the solvent **22** in the solvent housing **54**. The solvent **22** may be acetone or other type of organic solvent that is capable of dissolving extruded polystyrene thereby releasing air bubbles in the extruded polystyrene and the spray nozzle **70** may include a fluid pump for pumping the solvent **22** through the spray nozzle **70**.

[0030] The solvent bath unit **20** includes a tray ramp **65** that is coupled to and angles upwardly and rearwardly from the back wall **58** of the solvent housing **54**. Additionally, the tray ramp **65** is aligned with a tray opening **67** in the back wall **58** of the solvent housing **54** thereby facilitating a tray **69** which is placed on the tray ramp **65** to slide through the tray opening **67** and into the solvent housing **54** such that the chopped pieces **18** are deposited into the tray **69**. Additionally, the tray opening **67** is positioned beneath the entrance **64** in the solvent housing **54**. The tray **69** has a plurality of drain holes **71** each extending through a bottom wall **73** of the tray **69** thereby facilitating the solvent **22** to drain through each of the plurality of drain holes **71**. Furthermore, the tray **69** may be comprised of recycled plastic or other type of material that can be chopped or ground up by grinding blades.

[0031] The solvent bath unit **20** includes a solvent sensor **72** that is attached to the bottom wall **62** of the solvent housing **54** such that the solvent sensor **72** is positioned inside of the solvent housing **54**. The solvent **22** actuator is electrically coupled to the processing circuit **32** and the solvent sensor **72** senses when the tray **69** and the chopped pieces **18** are positioned in the solvent housing **54**. The solvent sensor **72** may comprise an optical sensor or other type of electronic sensor that is capable of sensing the presence of the tray **69** and the chopped pieces **18** in the solvent housing **54**.

[0032] The solvent bath unit **20** includes a solvent lift panel **80** which has a coupled end **82** and a free end **84** and the free end **84** of the solvent lift panel **80** is hingedly coupled to the bottom wall **62** of the solvent housing **54**. The solvent lift panel **80** is positionable in a first position having the solvent lift panel **80** lying on a plane that is oriented parallel to the bottom wall **62** of the solvent housing **54**. In this way the tray **69** and the chopped pieces **18** lie on the solvent lift panel **80** when the tray **69** and the chopped pieces **18** are submerged in the solvent **22**. The solvent lift panel **80** is positionable in a second position having the solvent lift panel **80** being oriented in a vertical position such that the solvent lift panel **80** pivots toward the exit **66** in the front wall **56** of the solvent housing **54**. In this way the solvent lift panel **80** can urge the tray **69** and the chopped pieces **18** through the exit **66**. Furthermore, the solvent **22** panel is foraminous thereby facilitating the solvent **22** to pass through the solvent lift panel **80** when the solvent **22** panel is urged into the second position for retaining the solvent **22** in the solvent housing **54**.

[0033] The solvent bath unit **20** includes a solvent lift panel actuator **86** which is attached to the solvent lift panel **80**. The solvent lift panel actuator **86** is electrically coupled to the processing circuit **32** and the solvent lift panel actuator **86** is actuated into a lifting condition when the processing circuit **32** receives the solvent lift input. Furthermore, the solvent lift panel **80** is tilted into the second position thereby facilitating the tray **69** and the chopped pieces **18** to be urged outwardly through the exit **66** in the front wall **56** when the solvent **22** lift actuator is actuated into the lifting condition. The solvent lift panel actuator **86** is actuated into a lowering condition when the solvent lift panel actuator **86** has been in the lifting condition for a sufficient duration of time to fully tilt the solvent lift panel **80** into the vertical position thereby positioning the solvent lift panel **80** in the first position. Additionally, the processing circuit **32** receives the solvent lift input when the solvent sensor **72** senses the tray **69** and the chopped pieces **18**.

[0034] The conveyor unit **24** comprises a conveyor pan **88** which has a forward end **90** and a rearward end **92** and a perimeter wall **94**. The rearward end **92** is coupled to the front wall **56** of the solvent housing **54** of the solvent **22** bath at a point that is aligned with the bottom wall **62** of the solvent housing **54**. The conveyor pan **88** is positioned beneath the exit **66** in the front wall **56** of the solvent housing **54**. The conveyor unit **24** includes a first conveyor belt **96** that is rotatably integrated into the conveyor pan **88** such that the first conveyor belt **96** extends between the forward end **90** and the rearward end **92** of the conveyor pan **88**. The tray **69** and the chopped pieces **18** are deposited onto the first conveyor belt **96** when the solvent lift panel actuator **86** tilts the solvent lift panel **80** into the vertical position.

[0035] The conveyor unit **24** includes a conveyor motor **98** which is attached to the conveyor pan **88** and the conveyor motor **98** is electrically coupled to the processing circuit **32**. The conveyor motor **98** is in mechanical communication with the first conveyor belt **96** such that the conveyor motor **98** urges the first conveyor belt **96** to travel between the rearward end **92** and the forward end **90** of the conveyor pan **88** thereby transporting the tray **69** and the chopped pieces **18** from the rearward end **92** of the conveyor pan **88** to the forward end **90** of the conveyor pan **88** when the conveyor motor **98** is turned on. The conveyor motor **98** is electrically coupled to the processing circuit **32** and the conveyor motor **98** is turned on when the processing circuit **32** receives the first conveyor input. Furthermore, the processing circuit **32** receives the first conveyor input when the processing circuit **32** receives the solvent lift input.

[0036] The freezing unit **26** comprises a freezer housing **111** which has a forward end **112**, a rearward end **113**, a lower wall **114** and an upper wall **115**; each of the forward end **112** and the rearward end **113** of the freezer housing **111** is open. The freezer housing **111** has a chute **116** extending rearwardly from the rearward end **113** of the freezer housing **111** at a point which is aligned with the lower wall **114** of the freezer housing **111**. Additionally, the chute **116** is attached to the forward end **90** of the conveyor pan **88** of the conveyor unit **24** such that the lower wall **114** of the freezer housing **111** lies on a plane that is coplanar with the basal wall **108** of the conveyor pan **88**. In this way the forward end **90** can receive the tray **69** and the chopped pieces **18** from the water lift panel **98**.

[0037] The freezing unit **26** includes a second conveyor belt **117** which is movably disposed on the lower wall **114** of the freezer housing **111** such that the second conveyor belt **117** extends substantially between the forward end **112** and the rearward end **113** of the freezer housing **111**. The second conveyor belt **117** is positioned inside of the freezer housing **111** such that the tray **69** and the chopped pieces **18** are deposited upon the second conveyor belt **117**. The second conveyor belt **117** includes a drive **118** which is electrically coupled to the processing circuit **32** and the drive **118** is actuated to rotate the second conveyor belt **117** when the processing circuit **32** receives the freezing actuate input. In this way the second conveyor belt **117** transports the tray **69** and the chopped pieces **18** from the rearward end **113** of the freezer housing **111** toward the forward end **112** of the freezer housing **111**. The drive **118** is de-actuated when the processing circuit **32** receives the freezing de-actuate input and the processing circuit **32** receives the heating de-actuate input after a sufficient duration of time for the second conveyor belt **117** to have transported the tray **69** and the chopped pieces **18** outwardly through the forward end **112** of the freezer housing **111**. Additionally, the duration of time for the second conveyor belt **117** to be actuated may range between approximately 5.0 minutes and 10.0 minutes.

[0038] The freezing unit **26** includes a freezer **119** that is integrated into the upper wall **115** of the freezer housing **111** such that the freezer **119** is in thermal communication with an interior of the freezer housing **111**. The freezer **119** is electrically coupled to the processing circuit **32** and the freezer **119** is turned on when the processing circuit **32** receives the freezing actuate input. Furthermore, the freezer **119** freezes the tray **69** and the chopped pieces **18** on the second conveyor belt **117** when the freezer **119** is turned on. In this way the solvent **22** and the tray **69** and the chopped pieces **18** are frozen for subsequent processing. The freezer **119** is turned off when the processing circuit **32** receives the freezing de-actuate input. Additionally, the freezer **119** may comprise an electric freezer which may have an operational temperature ranging between approximately −15.0 degrees Fahrenheit and 0.0 degrees Fahrenheit to facilitate the solvent and the tray and the chopped pieces to be freeze dried in a timely manner.

[0039] The grinding unit **28** comprises a duct **120** which has a back end **121** and a bottom end **122** and the duct **120** has a bend **123** which defines a first portion **124** of the duct **120** that is perpendicularly oriented with a second portion **125** of the duct **120**. The back end **121** is associated with the first portion **124** and the bottom end **122** is associated with the second portion **125**. Each of the back end **121** and the bottom end **122** is open and the duct **120** is flared adjacent to the bottom end **122** such that the bottom end **122** has dimensions that are greater than dimensions of the back end **121**. Additionally, the back end **121** is coupled to the forward end **112** of the freezer housing **111** of the freezing unit **26** such that the back end **121** receives the tray **69** and the chopped pieces **18** from the forward end **112** of the freezer housing **111**. Furthermore, the duct **120** is oriented such that the second portion **125** of the duct **120** extends downwardly thereby facilitating the tray **69** and the chopped pieces **18** the be gravity fed through the duct **120**.

[0040] The grinding unit **28** includes a plurality of grinding blades **126** and each of the plurality of grinding blades **126** is rotatably disposed in the second portion **125** of the duct **120**. Each of the plurality of grinding blades **126** is oriented to extend across the bottom end **122** of the duct **120** thereby forcing the tray **69** and the chopped pieces **18** to pass through the plurality of grinding blades **126** prior to passing through the bottom end **122** of the duct **120**. The plurality of grinding blades **126** may comprise cylindrical rollers that have a plurality of cutting edges or auger style screws or any other type of cutting device that is capable of grinding the tray **69** and the chopped pieces **18** into granules that have a maximum diameter ranging between approximately 1.5 mm and 3.0 mm.

[0041] The grinding unit **28** includes a blade drive **128** that is integrated into the second portion **125** of the duct **120**. The blade drive **128** is in mechanical communication with each of the plurality of grinding blades **126** such that the blade drive **128** rotates the plurality of grinding blades **126** when the blade drive **128** is turned on. In this way the plurality of grinding blades **126** can grind the tray **69** and the chopped pieces **18** into the granular material **30** having the granular material **30** exiting the duct **120** through the bottom end **122** of the duct **120**. Thus, the grinding blades **126** facilitate the granular material **30** to occupy as little space as possible in the waste management facility. The blade drive **128** is electrically coupled to the processing circuit **32** and the blade drive **128** is turned on when the processing circuit **32** receives the grinding input. Conversely, the blade drive **128** is turned off when the blade drive **128** has been turned on a sufficient duration of time to fully grind the tray **69** and the chopped pieces **18**.

[0042] A series of legs is **130** provided and each of the series of legs **130** is attached to and extends downwardly from the processing plant **12** thereby facilitating the processing plant **12** to be elevated above a support surface **131**. The support surface **131** may be a floor in an industrial facility or other type of horizontal support surface located in a facility that processes municipal waste, for example, or other type of facility that specializes in processing extruded polystyrene **16**. Each of the series of legs **130** has a telescopically adjustable length for adjusting the height and positioning of the processing plant **12** and each of the series of legs **130** includes a foot **132** that rests on the support surface **131**. The hopper **36** of the chopping unit **14** may be fed by a second conveyor belt, for example, or the hopper **36** of the chopping unit **14** may be manually fed by a worker. Additionally, the duct **120** of the grinding unit **28** may be directed toward a bed of a truck, for example, or other type of containment intended for transporting the granular material **30** to a waste management facility.

[0043] In use, the items of extruded polystyrene **16** are fed into the hopper **36** of the chopping unit **14** for processing the items of extruded polystyrene **16**. The chopping unit **14** chops the items of extruded polystyrene **16** into the chopped pieces **18** for subsequent reduction in the solvent bath unit **20**. Furthermore, a tray **69** is loaded onto the tray ramp **65** such that the tray **69** slides into the solvent bath unit **20** to facilitate the chopped pieces to be collected on the tray **69**. The tray **69** and the chopped pieces **18** are fed into the freezing unit **26** to be frozen and the tray **69** and the chopped pieces **18** are fed into the grinding unit **28** to grind the tray **69** and the chopped pieces **18** into the granular material **30** which can be collected and transported to a waste management facility, such a landfill for example. In this way the granular material **30** occupies far less space in the waste management facility than would the unprocessed items of extruded polystyrene **16**. Furthermore, the granular material **30** can be collected and employed for various recycling purposes or other means of reusing the polystyrene.

[0044] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, device and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0045] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word “comprising” is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article “a” does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

Claims

1. An extruded polystyrene processing device for reducing extruded polystyrene into a granular material for subsequent delivery to a waste management facility, said device comprising: a processing plant including a chopping unit for chopping extruded polystyrene into chopped pieces, said chopping unit being connected to a solvent bath unit for submerging said chopped pieces in a solvent for eliminating air bubbles in said chopped pieces, said solvent bath unit being connected to a conveyor unit which receives said chopped pieces for transporting the solvent from said chopped pieces, said conveyor unit being connected to a freezing unit which receives said chopped pieces from said conveyor unit to subsequently freeze said chopped pieces, said freezing unit being connected to a grinding unit which receives said frozen chopped pieces for grinding said frozen chopped pieces into a granular material for subsequent collection and transportation to a waste management facility wherein said processing plant is configured to reduce the total volume of the extruded polystyrene prior to being handled by the waste management facility thereby reducing the environmental impact of the extruded polystyrene.

2. The device according to claim 1, wherein: said processing plant includes a processing circuit; said processing circuit receives a chop input; said processing circuit receives a door open input; said processing circuit receives a door close input; said processing circuit receives a solvent lift input; said processing circuit receives a first conveyor input; said processing circuit receives a second conveyor input; said processing circuit receives a freezing actuate input; said processing circuit receives a freezing de-actuate input; said processor receives a grinding input; and said processing circuit is electrically coupled to a power source.

3. The device according to claim 1, wherein: said chopping unit comprises a hopper having a first end and a second end and an outer wall extending between said first end and said second end; said hopper is substantially hollow; each of said first end and said second end is open; and said outer wall undulates between said first end and said second end such that said hopper defines an S-shape having said first end being spaced upwardly from said second end wherein said first end is configured to receive items of extruded polystyrene such that said undulation of said outer wall facilitates the items of extruded polystyrene to be gravity fed toward said second end.

4. The device according to claim 3, wherein: said processing plant includes a processing circuit; said processing circuit receives a chop input; said chopping unit includes a chopper being movably integrated within said hopper; said chopper is electrically coupled to said processing circuit; said chopper is turned on when said processing circuit receives said chop input; said chopper is positioned closer to said first end than said second end; and said chopper includes a series of blades which are rotated when said chopper is turned on thereby facilitating said chopper to chop the items of extruded polystyrene into said chopped pieces.

5. The device according to claim 4, wherein: said chopping unit includes a sensor being positioned within said hopper; said sensor is positioned closer to said second end of said hopper than said chopper; said sensor is electrically coupled to said processing circuit; said sensor senses the volume of chopped pieces in said hopper; said processing circuit receives a door open input when said sensor determines that the volume of said chopped pieces in said hopper exceeds a pre-determined trigger volume; said pre-determined trigger volume is a volume of approximately 80.0 percent of the internal area in said hopper which is located between said chopper and said second end of said hopper; and said processing circuit receives a door close input when said sensor determines that the volume of said chopped pieces in said hopper does not exceed said pre-determined trigger volume.

6. The device according to claim 3, wherein: said chopping unit includes a door being slidably integrated into said second end of said hopper; said door is oriented to lie on a plane being oriented parallel to said second end of said hopper; said door is positionable in a closed position having said door lying against said second end for closing said second end thereby inhibiting said chopped pieces from passing through said second end and exiting said hopper; and said door is positionable in an open position having said door being lifted upwardly to expose said second end thereby facilitating said chopped pieces to pass through said second end to exit said hopper.

7. The device according to claim 6, wherein: said processing plant includes a processing circuit; said processing circuit receives a door close input and a door open input; and said chopping unit includes a door actuator being integrated into said hopper; said door actuator is electrically coupled to said processing circuit; said door actuator is in mechanical communication with said door; said door actuator is actuated into a closing condition when said processing circuit receives said door close input thereby urging said door into said closed position; said door actuator is actuated into an open condition when said processing circuit receives said door open input thereby urging said door into said open position.

8. The device according to claim 6, wherein: said solvent bath unit comprises a solvent housing having a front wall and a back wall and a top wall and a bottom wall; said solvent housing has an entrance extending through said back wall; said solvent housing has an exit extending through said front wall; each of said exit and said entrance is spaced upwardly from said bottom wall; said second end of said hopper is attached to said back wall of said solvent housing having said second end being aligned with said entrance thereby facilitating said chopped pieces to pass through said entrance when said door is urged into said open position; said solvent housing has a drain being integrated into said bottom wall thereby facilitating said drain to drain a solvent contained in said solvent housing; said solvent is filled to a level that is below each of said entrance and said exit in said solvent housing thereby facilitating said solvent housing to contain said solvent; said solvent housing has a spray nozzle extending through said top wall of said solvent housing; and said spray nozzle is in fluid communication with said drain thereby facilitating said spray nozzle to spray said solvent into said solvent housing for recirculating said solvent in said solvent housing.

9. The device according to claim 8, wherein: said processing plant includes a processing circuit; said solvent bath unit includes a tray ramp being coupled to and angling upwardly and rearwardly from said back wall of said solvent housing; said tray ramp is aligned with a tray opening in said back wall of said solvent housing thereby facilitating a tray which is placed on said tray ramp to slide through said tray opening and into said solvent housing such that said chopped pieces are deposited into said tray; said tray has a plurality of drain holes each extending through a bottom wall of said tray thereby facilitating said solvent to drain through each of said plurality of drain holes; said solvent bath unit includes a solvent sensor being attached to said bottom wall of said solvent housing such that said solvent sensor is positioned inside of said solvent housing; and said

solvent sensor senses when said tray and said chopped pieces are positioned in said solvent housing.
10. The device according to claim 8, wherein: said solvent bath unit includes a solvent lift panel having a coupled end and a free end; said free end of said solvent lift panel is hingedly coupled to said bottom wall of said solvent housing; said solvent lift panel is positionable in a first position having said solvent lift panel lying on a plane being oriented parallel to said bottom wall of said solvent housing thereby facilitating said tray and said chopped pieces to lie on said solvent lift panel when said tray and said chopped pieces are submerged in said solvent; said solvent lift panel has a foot extending upwardly from said solvent lift panel at a point being aligned with said free end of said solvent lift panel for retaining said tray on said solvent lift panel; said solvent lift panel is positionable in a second position having said solvent lift panel being oriented in a vertical position having said solvent lift panel pivoting toward said exit in said front wall of said solvent housing thereby facilitating said solvent lift panel to urge said tray and said chopped pieces through said exit; and said solvent panel is foraminous thereby facilitating said solvent to pass through said solvent lift panel when said solvent panel is urged into said second position for retaining said solvent in said solvent housing.
11. The device according to claim 10, wherein: said processing plant includes a processing circuit; said processing circuit receives a solvent lift input; said solvent bath includes a solvent lift panel actuator being attached to said solvent lift panel; said solvent lift panel actuator is electrically coupled to said processing circuit; said solvent lift panel actuator is actuated into a lifting condition when said processing circuit receives said solvent lift input; said processing circuit receiving said solvent lift input when said solvent sensor senses said tray and said chopped pieces; said solvent lift panel has a foot extending upwardly from said solvent lift panel at a point being aligned with said free end of said solvent lift panel for retaining said tray on said solvent lift panel; said solvent lift panel is tilted into said second position thereby facilitating said tray and said chopped pieces to be urged outwardly through said exit in said front wall when said solvent lift actuator is actuated into said lifting condition; and said solvent lift panel actuator is actuated into a lowering condition when said solvent lift panel actuator has been in said lifting condition for a sufficient duration of time to fully tilt said solvent lift panel into said vertical position thereby positioning said solvent lift panel in said first position.
12. The device according to claim 9, wherein: said conveyor unit comprises a pan having a forward end and a rearward end and a perimeter wall; said rearward end is coupled to said front wall of said solvent housing of said solvent bath at point being aligned with said bottom wall of said solvent housing having said pan being positioned beneath said exit in said front wall of said solvent housing; said conveyor unit includes a first conveyor belt being rotatably integrated into said conveyor pan such that said first conveyor belt extends between said forward end and said rearward end of said conveyor pan, said tray being deposited onto said first conveyor belt when said solvent lift actuator tilts said solvent lift panel into said vertical position; said conveyor unit includes a conveyor motor being attached to said conveyor pan; said conveyor motor is electrically coupled to said processing circuit; said processing circuit receiving a solvent lift input; said processing circuit receiving a first conveyor input; said conveyor motor is in mechanical communication with said first conveyor belt such that said conveyor motor urges said conveyor belt to travel between said rearward end and said forward end of said conveyor pan thereby transporting said tray from said rearward end of said conveyor pan to said forward end of said conveyor pan when said conveyor motor is turned on; said conveyor motor is turned on when said processing circuit receives said first conveyor input; and said processing circuit receives said first conveyor input when said processing circuit receives said solvent lift input.
13. The device according to claim 12, wherein: said processing circuit receives a freezing actuate input and a freezing de-actuate input; said freezer unit includes a freezer housing having a forward end and a rearward end and a lower wall and an upper wall; each of said forward end and said

rearward end of said freezer housing is open; said rearward end of said freezer housing is attached to said forward end of said conveyor pan such that said rearward end of said freezer housing receives said tray from said first conveyor belt; said freezing unit includes a second conveyor belt being movably disposed on said lower wall of said freezer housing such that said second conveyor belt extends substantially between said forward end and said rearward end of said freezer housing; said second conveyor belt is positioned inside of said freezer housing such that said chopped pieces are deposited upon said second conveyor belt; said second conveyor belt includes a drive being electrically coupled to said processing circuit; said drive is actuated to rotate said second conveyor belt when said processing circuit receives said freezing actuate input thereby facilitating said second conveyor belt to transport tray and said chopped pieces from said rearward end of said freezer housing toward said forward end of said freezer housing; said drive is de-actuated when said processing circuit receives said freezing de-actuate input; and said processing circuit receives said heating de-actuate input after a sufficient duration of time for said second conveyor belt to have transported said tray and said chopped pieces outwardly through said forward side of said freezer housing.

14. The device according to claim 13, wherein: said freezing unit includes a freezer being integrated into said upper wall of said freezer housing such that said freezer is in thermal communication with an interior of said freezer housing; said freezer is electrically coupled to said processing circuit; said freezer is turned on when said processing circuit receives said freezing actuate input; said freezer freezes said tray and said chopped pieces on said second conveyor belt when said freezer is turned on for freezing said water that remains on said tray and said chopped pieces; and said freezer is turned off when said processing circuit receives said freezing de-actuate input.

15. The device according to claim 13, wherein: said grinding unit comprises a duct having a back end and a bottom end; said duct has a bend defining a first portion of said duct being perpendicularly oriented with a second portion of said duct; said back end is associated with said first portion; said bottom end being associated with said second portion; each of said back end and said bottom end is open; said duct is flared adjacent to said bottom end such that said bottom end has dimensions being greater than dimensions of said back end; said back end is coupled to said forward end of said freezer housing of said freezing unit such that said back end receives said tray and said chopped pieces from said forward end of said freezer housing; and said duct is oriented such that said second portion of said duct extends downwardly thereby facilitating said tray and said chopped pieces the be gravity fed through said duct.

16. The device according to claim 15, wherein: said processing circuit receives a grinding input; said grinding unit includes a plurality of grinding blades; each of said plurality of grinding blades is rotatably disposed in said second portion of said duct; each of said plurality of grinding blades is oriented to extend across said bottom end of said duct thereby forcing said tray and said chopped pieces to pass through said plurality of blades prior to passing through said bottom end of said duct; said grinding unit includes a blade drive being integrated into said second portion of said duct; said blade drive is in mechanical communication with each of said plurality of grinding blades such that said blade drive rotates said plurality of grinding blades when said blade drive is turned on thereby facilitating said plurality of grinding blades to grind said tray and said chopped pieces into granular material having the granular material exiting said duct through said bottom end of said duct wherein said grinding blades are configured to facilitate the granular material to occupy as little space as possible in the waste management facility; and said blade drive is electrically coupled to said processing circuit, said blade drive being turned on when said processing circuit receives said grinding input, said blade drive being turned off when said blade drive has been turned on a sufficient duration of time to fully grind said chopped pieces.

17. An extruded polystyrene processing device for reducing extruded polystyrene into a granular material for subsequent delivery to a waste management facility, said device comprising: a processing plant including a chopping unit for chopping extruded polystyrene into chopped pieces, said chopping unit being connected to a solvent bath unit for submerging said chopped pieces in a solvent for eliminating air bubbles in said chopped pieces, said solvent bath unit being connected to a conveyor unit which receives said chopped pieces for transporting the solvent from said chopped pieces, said conveyor unit being connected to a freezing unit which receives said chopped pieces from said conveyor unit to subsequently freeze and dry said chopped pieces, said freezing unit being connected to a grinding unit which receives the frozen and dried chopped pieces for grinding the frozen and dried chopped pieces into a granular material for subsequent collection and transportation to a waste management facility wherein said processing plant is configured to reduce the total volume of the extruded polystyrene prior to being handled by the waste management facility thereby reducing the environmental impact of the extruded polystyrene; wherein said processing plant includes a processing circuit, said processing circuit receiving a chop input, said processing circuit receiving a door open input, said processing circuit receiving a door close input, said processor receiving a solvent lift input, said processing circuit receiving a first conveyor input, said processing circuit receiving a second conveyor input, said processing circuit receiving a freezing actuate input, said processing circuit receiving a freezing de-actuate input, said processor receiving a grinding input, said processing circuit being electrically coupled to a power source; wherein said chopping unit comprises: a hopper having a first end and a second end and an outer wall extending between said first end and said second end, said hopper being substantially hollow, each of said first end and said second end being open, said outer wall undulating between said first end and said second end such that said hopper defines an S-shape having said first end being spaced upwardly from said second end wherein said first end is configured to receive items of extruded polystyrene such that said undulation of said outer wall facilitates the items of extruded polystyrene to be gravity fed toward said second end; a chopper being movably integrated within said hopper, said chopper being electrically coupled to said processing circuit, said chopper being turned on when said processing circuit receives said chop input, said chopper being positioned closer to said first end than said second end, said chopper including a series of blades which are rotated when said chopper is turned on thereby facilitating said chopper to chop the items of extruded polystyrene into said chopped pieces; a sensor being positioned within said hopper, said sensor being positioned closer to said second end of said hopper than said chopper, said sensor being electrically coupled to said processing circuit, said sensor sensing the volume of chopped pieces in said hopper, said processing circuit receiving said door open input when said sensor determines that the volume of said chopped pieces in said hopper exceeds a pre-determined trigger volume, said pre-determined trigger volume being a volume of approximately 80.0 percent of the internal area in said hopper which is located between said chopper and said second end of said hopper, said processing circuit receiving said door close input when said sensor determines that the volume of said chopped pieces in said hopper does not exceed said pre-determined trigger volume; a door being slidably integrated into said second end of said hopper, said door being oriented to lie on a plane being oriented parallel to said second end of said hopper, said door being positionable in a closed position having said door lying against said second end for closing said second end thereby inhibiting said chopped pieces from passing through said second end and exiting said hopper, said door being positionable in an open position having said door being lifted upwardly to expose said second end thereby facilitating said chopped pieces to pass through said second end to exit said hopper; and a door actuator being integrated into said hopper, said door actuator being electrically coupled to said processing circuit, said door actuator being in mechanical communication with said door, said door actuator being actuated into a closing condition when said processing circuit receives said door close input thereby urging said door into said closed position, said door actuator being actuated into an open condition when said processing circuit receives said door open input thereby urging said door into said open position; wherein said solvent bath comprises: a solvent housing having a front wall and a back wall and a top wall and a bottom wall, said solvent housing having an entrance extending through said back wall, said solvent housing having an exit extending through said front wall, each of said exit and said entrance being spaced upwardly from said bottom wall, said second end of said hopper being attached to said back wall of said solvent housing having said second end being aligned with said entrance thereby facilitating said chopped pieces to pass through said entrance when said door is urged into said open position, said solvent housing having a drain being integrated into said bottom wall thereby facilitating said drain to drain a solvent contained in said solvent housing, said solvent being filled to a level that is below each of said entrance and said exit in said solvent housing thereby facilitating said solvent housing to contain said solvent, said solvent housing having a spray nozzle extending through said top wall of said solvent housing, said spray nozzle being in fluid communication with said drain thereby facilitating said spray nozzle to spray said solvent into said solvent housing for recirculating said solvent in said solvent housing; a tray ramp being coupled to and angling upwardly and rearwardly from said back wall of said solvent housing, said tray ramp being aligned with a tray opening in said back wall of said solvent housing thereby facilitating a tray which is placed on said tray ramp to slide through said tray opening and into said solvent housing such that said chopped pieces are deposited into said tray, said tray having a plurality of drain holes each extending through a bottom wall of said tray thereby facilitating said solvent to drain through each of said plurality of drain holes; a solvent sensor being attached to said bottom wall of said solvent housing such that said solvent sensor is positioned inside of said solvent housing, said solvent actuator being electrically coupled to said processing circuit, said solvent sensor sensing when said chopped pieces are positioned in said solvent housing; a solvent lift panel having a coupled end and a free end, said free end of said solvent lift panel being hingedly coupled to said bottom wall of said solvent housing, said solvent lift panel being positionable in a first position having said solvent lift panel lying on a plane being oriented parallel to said bottom wall of said solvent housing thereby facilitating said tray to rest on said solvent lift panel when said tray is slid into said solvent housing thereby submerging said tray and said chopped pieces in said solvent, said solvent lift panel having a foot extending upwardly from said solvent lift panel at a point being aligned with said free end of said solvent lift panel for retaining said tray on said solvent lift panel, said solvent lift panel being positionable in a second position having said solvent lift panel being oriented in a vertical position having said solvent lift panel pivoting toward said exit in said front wall of said solvent housing thereby facilitating said solvent lift panel to urge said tray and said chopped pieces through said exit, said solvent panel being foraminous thereby facilitating said solvent to pass through said solvent lift panel when said solvent panel is urged into said second position for retaining said solvent in said solvent housing; and a solvent lift panel actuator being attached to said solvent lift panel, said solvent lift panel actuator being electrically coupled to said processing circuit, said solvent lift panel actuator being actuated into a lifting condition when said processing circuit receives said solvent lift input, said processing circuit receiving said solvent lift input when said solvent sensor senses said tray and said chopped pieces, said solvent lift panel being tilted into said second position thereby facilitating said tray to be urged outwardly through said exit in said front wall when said solvent lift actuator is actuated into said lifting condition, said solvent lift panel actuator being actuated into a lowering condition when said solvent lift panel actuator has been in said lifting condition for a sufficient duration of time to fully tilt said solvent lift panel into said vertical position thereby positioning said solvent lift panel in said first position; said conveyor unit comprises: a conveyor pan having a forward end and a rearward end, said rearward end is coupled to said front wall of said solvent housing of said solvent bath at point being aligned with said bottom wall of said solvent housing having said pan being positioned beneath said exit in said front wall of said solvent housing; a first conveyor belt being rotatably integrated into said conveyor pan such that said first conveyor belt extends between said forward end and said rearward end of said conveyor pan, said tray being deposited onto said first conveyor belt when said solvent lift actuator tilts said solvent lift panel into said vertical position; a conveyor motor being attached to said conveyor pan, said conveyor motor being electrically coupled to said processing circuit, said conveyor motor being in mechanical communication with said first conveyor belt such that said conveyor motor urges said conveyor belt to travel between said rearward end and said forward end of said conveyor pan thereby transporting said tray from said rearward end of said conveyor pan to said forward end of said conveyor pan when said conveyor motor is turned on, said conveyor motor being electrically coupled to said processing circuit, said conveyor motor being turned on when said processing circuit receives said first conveyor input, said processing circuit receiving said first conveyor input when said processing circuit receives said solvent lift input; wherein said freezing unit comprises: a freezer housing having a forward end and a rearward end and a lower wall and an upper wall, each of said forward end and said rearward end of said freezer housing being open, said rearward end of said freezer housing being attached to said forward end of said conveyor pan such that said rearward end of said freezer housing receives said tray and said chopped pieces from said first conveyor belt; a second conveyor belt being movably disposed on said lower wall of said freezer housing such that said second conveyor belt extends substantially between said forward end and said rearward end of said freezer housing, said second conveyor belt being positioned inside of said freezer housing such that said tray and said chopped pieces are deposited upon said second conveyor belt, said second conveyor belt including a drive being electrically coupled to said processing circuit, said drive being actuated to rotate said second conveyor belt when said processing circuit receives said freezing actuate input thereby facilitating said second conveyor belt to transport said tray and said chopped pieces from said rearward end of said freezer housing toward said forward end of said freezer housing, said drive being de-actuated when said processing circuit receives said freezing de-actuate input, said processing circuit receiving said heating de-actuate input after a sufficient duration of time for said second conveyor belt to have transported said tray and said chopped pieces outwardly through said forward side of said freezer housing; and a freezer being integrated into said upper wall of said freezer housing such that said freezer is in thermal communication with an interior of said freezer housing, said freezer being electrically coupled to said processing circuit, said freezer being turned on when said processing circuit receives said freezing actuate input, said freezer freezing said tray and said chopped pieces on said second conveyor belt when said freezer is turned on for freezing said water that remains on said tray and said chopped pieces, said freezer being turned off when said processing circuit receives said freezing de-actuate input; and wherein said grinding unit comprises: a duct having a back end and a bottom end, said duct having a bend defining a first portion of said duct being perpendicularly oriented with a second portion of said duct, said back end being associated with said first portion, said bottom end being associated with said second portion, each of said back end and said bottom end being open, said duct being flared adjacent to said bottom end such that said bottom end has dimensions being greater than dimensions of said back end, said back end being coupled to said forward end of said freezer housing of said freezing unit such that said back end receives said tray and said chopped pieces from said forward end of said freezer housing, said duct being oriented such that said second portion of said duct extends downwardly thereby facilitating said tray and said chopped pieces the be gravity fed through said duct; a plurality of grinding blades, each of said plurality of grinding blades being rotatably disposed in said second portion of said duct, each of said plurality of grinding blades being oriented to extend across said bottom end of said duct thereby forcing said tray and said chopped pieces to pass through said plurality of blades prior to passing through said bottom end of said duct; and a blade drive being integrated into said second portion of said duct, said blade drive being in mechanical communication with each of said plurality of grinding blades such that said blade drive rotates said plurality of grinding blades when said blade drive is turned on thereby facilitating said plurality of grinding blades to grind said tray and said chopped pieces into a granular material having said granular material exiting said duct through said bottom end of said duct wherein said grinding blades are configured to facilitate said granular material to occupy as little space as possible in the waste management facility, said blade drive being electrically coupled to said processing circuit, said blade drive being turned on when said processing circuit receives said grinding input, said blade drive being turned off when said blade drive has been turned on a sufficient duration of time to fully grind said tray and said chopped pieces.